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EFFECT OF CAPTIONS AND POLITICAL IDEOLOGY ON INITIAL EMOTIONAL RESPONSE TO IMAGES

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Abstract – In the current media climate, news items are presented in a way to grab attention and clicks by connecting with the public's attitudes. There are multiple ways in which this is achieved. This study focuses on how the media changes the verbal message associated with images to be relevant to their audience and agenda. The manipulation of the verbal framing feeds the political ideology of the media outlet's target audience. These ideological differences could elicit different emotional responses. The current study explored how image captions and political ideology predict the emotional response to news images. We distributed an online script to a convenience sample of undergraduate university students, then obtained a Psi Chi grant to fund the acquisition of a more representative sample from across the United States. We found that different captions associated with a single image induced different emotional responses, depending on the image. The generalizability between the convenience and community samples was modest, but future studies that redress the limitations of our experiment will contribute to the discussion of whether convenience samples generalize to the overall population.

Keywords: captions, emotional response, political ideology, political attitudes, image processing

In past decades, American news broadcasts were dominated by the three major television networks, but the rapid proliferation of news outlets in more recent years has fragmented the market to the point that each outlet needs to cater to a narrower slice of the audience. In today's news climate, several trends are apparent: the use of images, different stories about the same events, and stories catering to certain audiences. We wanted to see if we could manipulate how people respond to politically charged images by varying the captions associated with the images. By doing so we hoped to study a phenomenon that is commonly observed in the media's presentation of news: two different media outlets can present the same video footage, but the verbal description of the video varies such that it conforms to each outlet's political stance.

The Role of Cognition

Through everyday experiences, people create and revise structures in long-term memory called schemas that contain their overall knowledge on particular topics. These schemas help people to recognize common objects, behave appropriately in various situations, and predict what is likely to happen in those situations filled with those objects (Axelsson, 2007). Allen and Allison (2016) describe two processes, conscious and unconscious, that help explain how schemas are created and how the responses to the schemas occur. The first process is effortful, limited in capacity, and consciously mediated, and the second is unconscious and automatic. These two processes correspond to Kahneman's well-known distinction between system one and system two, respectively (Kahneman, 2011). System one determines when new information should be transferred to the conscious mind to make sense of it. When a stimulus is changed, it can change the schema that is retrieved from memory. In a study on jury trial reactions, when the testimony was changed to elicit a more emotional response, there was a great change in the reaction (Salerno, 2017). In another study (Bertrams et al., 2020), presenting feedback in a red font resulted in a temporary decline in cognitive performance relative to feedback presented in a neutral color.

These and other studies suggest that emotion affects the cognitive and behavioral reactions to stimuli. Yang et al. (2012) argued that emotional processing operates faster than cognitive functioning, and Kim and

Oh (2020) suggest that emotional responses trigger changes in judgment and behavior. Because emotion is processed faster it lays the foundation of our thoughts, and could influence the way someone interprets a situation.

Political Ideology

Political ideology is a great influence on people's thoughts and behaviors in today's society (Habib et al., 2020). Emotion and political ideology work side by side to affect cognition and behavior (Aubin, 1996). This is the stance that the current study will be evaluating. McHugo et al. (1985) mentions how political identification and beliefs play a role in emotional reactions to news and political leaders. Participants' political attitudes prior to seeing an image of a political leader expressing an emotion affected their reaction to the image of the political leader. When participants identified with a similar attitude as the political leader they showed more attitudes of warmth and joy regardless of the emotional display of the political leader. This is consistent with Habib et al.'s (2020) argument that people are motivated by evidence that supports their view over opposing views. Applying this principle to political ideology, when confronted by an issue that supports one's own political perspective, the individual feels happier and more supported than when confronted by an issue that does not support that political perspective, and which could cause negative emotions like anger.

Liberals and conservatives differ in their views, and tend to use different language to describe their perspectives and experiences. Conservatives tend to be conscientious and desire order, whereas liberals tend to be more open to new experiences (Laverghetta et al., 2007). Liberals prefer words like "triumph, innovation and privilege" while conservatives prefer words like "authority, security and law" to describe different social issues (Sterling et al., 2020). These differences in values that are shown through language could play a role in why news networks use certain diction and tactics to draw in the attention of the viewer. Partisans from both sides of the political spectrum see neutral news as biased (Emily & Vraga, 2015). This suggests the reason that media outlets have an incentive to slant their coverage to match the values of their target viewers (Emily & Vraga, 2015). We hoped to study a phenomenon that is commonly observed in the media's presentation of news: two different media outlets can present the same video footage, but the verbal description of the footage varies between the outlets to conform to each outlet's political orientation.

In our experiment we selected three politically charged images: one depicting people wearing masks in the time of COVID, a second depicting anti-abortion protestors, and a third depicting a confrontation between Black Lives Matter protesters and police. The images appeared either by themselves or accompanied by a caption, and when a caption was present, it was slanted toward a liberal or conservative viewpoint. Thus, the independent variable was the type of caption, with three levels: no caption, liberal caption, and conservative caption. After viewing each image, participants reported their emotional state; this was the dependent variable. Finally, after participants viewed the three images, they completed a political attitudes inventory.

We had two hypotheses. First, we expected liberal and conservative captions to elicit different emotional responses, and the emotional responses for images without captions to lie between the extremes staked out by the liberal and conservative captions. Second, we expected the congruity between political ideology and the caption to drive an emotional response, such that participants with a relatively liberal ideology should have a more positively valenced emotional reaction to liberal captions than to conservative captions, and participants with a relatively conservative ideology should have a more positively valenced emotional reaction to conservative captions than to liberal captions.

Method

Participants

We submitted an application for review to our university's Institutional Review Board entitled Effects of Descriptive Captions on Initial Emotional Response to Images: Political Ideology Predictors. After the IRB informed us that our application was accepted, we began to collect data, and treated all participants in accordance with the ethical guidelines established by the American Psychological Association (2017).

We distributed an online questionnaire to two samples of participants. The first sample was drawn from the undergraduate student population at a mid-sized university in the southwestern United States, who received course credit in exchange for their participation. Although it is convenient to sample from undergraduate students who have an experimental participation requirement in various psychology courses, the results may not generalize to the overall population (Hanel & Vione, 2016; Peterson & Marunka, 2014). For that reason, after we gathered data from undergraduate students, we wanted a more representative sample. To that end, the first author applied for an undergraduate research grant from Psi Chi, the international honor society in psychology. With monetary support from the Psi Chi Undergraduate Research Grant, we paid Qualtrics to recruit a diverse sample of participants from across the

United States. We did not specify any constraints on the demographics of the sample, except that all participants should be at least 18 years old.

The first sample consisted of 205 undergraduate students enrolled in psychology courses between the ages of 17 and 51 (M = 20.28, SD = 4.13); 157 identified as female, 42 as male, two as nonbinary, two as transgender, and two preferred not to report their gender. In response to the question about race, 142 participants described themselves as White, 28 as Black, 16 as mixed race, eight as Hispanic, seven as Asian, and four preferred not to report their race. The second sample consisted of 350 participants from across the country between the ages of 18 and 100 (M = 38.44, SD = 12.94); 191 identified as female, 155 as male, one as transgender female, one as genderfluid transmasc and two preferred not to respond. In terms of race, 225 described themselves as White, 48 as Black, 25 as mixed race, 24 as Asian, 22 as Hispanic, three as American Indian or Alaska Native, two as Native Hawaiian, and one preferred not to respond.

Materials

Script

Participants used their own electronic devices to access a script we wrote and posted on Qualtrics.com. The undergraduate students connected to their accounts on Sona (ourUniversity.sona-systems.com), which redirected them to the Qualtrics script. When participants completed the script, Qualtrics directed Sona to grant them credit for their participation. To coordinate Sona accounts with the Qualtrics script, we used instructions available at the Sona website. For the nationwide sample, we paid Qualtrics with the funds from the Psi Chi grant, which Qualtrics used to recruit and pay participants.

Images and Captions

We found three politically charged images on the internet: one depicts people in public wearing masks, the second depicts antiabortion protestors, and the third depicts Black Lives Matter (BLM) protestors confronting police. Participants viewed the three images in that order, with images of abstract art between every two political images, to provide a break from any intense emotional arousal the politically charged images may have induced. Each political image was presented either alone or with a caption. Based on a study of linguistics of political ideologies (Sterling et al., 2020), we wrote the captions to express either a liberal (e.g., U.S. citizens choose to wear masks for public safety) or conservative (e.g., U.S. citizens are ordered to wear face masks for COVID protection) perspective. We identified language used amongst conservative and liberal Twitter users found within this study and paired them with extreme views of conservative and liberal politics. The images and associated captions in each of three between-subjects conditions are depicted in Appendix A.

Scales

Positive and Negative Affect Scale. To measure emotional reaction to the images, participants completed a revised version of the Positive and Negative Affect Scale (PANAS; Watson, 1988). From the 20 emotions in the PANAS, we selected five pairs of words that express opposing emotional valence: Disgusted-Interested, Ashamed-Proud, Upset-Inspired, Jittery-Strong, and Irritable-Enthusiastic. Below each image (and caption if there was one) was a series of five sliders that could each be moved horizontally to any of five positions from -2 to +2. The -2 position for each slider was labeled with that slider's negative valence emotion, the 0 position was labeled with "Neutral", and the +2 position was labeled with that slider's positive valence emotion. All five sliders were initially positioned at "Neutral."

Ideological Consistencies Scale. To measure political ideology, participants completed the Ideological Consistency Scale (ICS; Pew Research, 2015). Questions in the ICS are organized in pairs, with the first in each pair being consistent with a conservative viewpoint (e.g., Government is almost always wasteful) and the second consistent with a liberal viewpoint (e.g., Government often does a better job than people give it credit for). Participants reported their reaction to each statement by clicking on one of five radio buttons with the following labels: Strongly disagree, Disagree, Neutral, Agree, and Strongly agree. All statements for this scale appear in Appendix B.

Procedure

The Qualtrics script began by presenting an informed consent letter. In the bottom right corner of each Qualtrics page is a button emblazoned with an arrow pointing to the right that participants click when they want to proceed to the next page. After participants advanced from the informed consent letter, they were presented with a page containing demographics questions. Following the demographics questionnaire was a series of five pages presented one at a time. Each page contained one image and a PANAS questionnaire. The first, third, and fifth pages contained the mask, abortion protest, and BLM images, respectively, and the second and fourth pages contained images of abstract art. Participants were randomly assigned to one of three conditions that dictated whether each image was accompanied by a caption, and what the caption was, as depicted in Appendix A. Participants could advance to the

next page only after completing the condensed version of the PANAS by positioning the five slider controls to represent their current affective state. After proceeding through the five pages that each contained an image and PANAS with slider controls, all participants completed the Ideological Consistency Scale. Finally, a debriefing message described the purpose of the study and our hypotheses, and provided contact information if participants had any questions.

Results

We planned to analyze the data as follows. Mean PANAS responses were submitted to a 3 x 3 ANOVA with Image as a within-participants factor and Caption as a between-participants factor. To see if the results supported the first hypothesis that the caption should influence PANAS responses, planned contrasts examined the difference between the liberal and conservative captions.

The second hypothesis predicted that participants with a relatively liberal ideology should have a more positively valenced emotional reaction to liberal captions than to conservative captions, and participants with a relatively conservative ideology should have a more negatively valenced emotional reaction to liberal captions than to conservative captions. The first step in testing the second hypothesis was to calculate an ICS score for each participant. The five responses from "strongly disagree" to "strongly agree" were assigned the numbers from one to five, respectively, for the conservative statements (e.g., Government is almost always wasteful and inefficient), and reverse coded from five to one, respectively, for the liberal statements (e.g., Government often does a better job than people give it credit for). The sum of all responses to the twenty statements yielded an overall score for each participant. As a result, participants with relatively liberal attitudes should have numerically smaller ICS scores than participants with relatively conservative attitudes. We submitted the ICS scores to an independent samples t-test with sample (undergraduate versus nationwide) as the grouping variable to determine if the undergraduate student sample was more liberal or conservative than the nationwide sample.

For participants who viewed a conservative caption for one of the images, our second hypothesis predicts that liberal participants (i.e., those with numerically low ICS scores) should have a lower PANAS score than conservative participants (i.e., numerically high ICS scores). In other words, the correlation between ICS and PANAS should be positive. On the other hand, for participants who viewed a liberal caption, liberal participants (low ICS) should have a higher PANAS than conservative participants (high ICS), which should yield a negative correlation between ICS and PANAS. Thus, our second hypothesis predicts that the correlation coefficient for ICS and PANAS should be higher for conservative captions than for liberal captions.

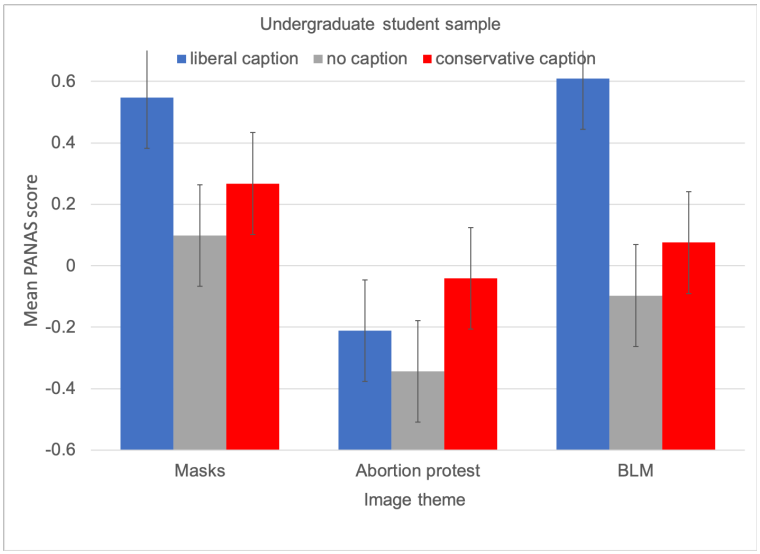
To determine if our results supported the second hypothesis, for each of the three images we calculated correlation coefficients for participants who viewed the conservative caption for that image and participants who viewed the liberal caption for that image. Then we submitted the two correlation coefficients to a Fisher's r to z transformation. A significant z in the right tail would support our second hypothesis.

Undergraduate Student Sample

PANAS. The internal consistency of the PANAS was acceptable ($\propto$ = .69). Mean PANAS scores (depicted in Figure 1) were submitted to a 3 x 3 ANOVA with Image as a within-participants factor and Caption as a between-participants factor. The main effect of Image, $F(2, 404) = 10.25$, $p < .001$, $\eta_p^2 = .048$, and the interaction between Image and Caption, $F(4, 404) = 4.36$, $p = .002$, $\eta_p^2 = .041$, were significant, but the main effect of Caption, $F(2, 202) = 2.1$, $p = .12$, $\eta_p^2 = .021$, was not. As can be seen in Figure 1, the effect of Image appears to be driven by higher

Figure 1

Mean PANAS responses as a function of Image and Caption for undergraduate student sample



Note. Error bars represent 95% confidence intervals, calculated as described in Loftus & Masson (1994).

PANAS values for the mask and BLM images than for the abortion protest image.

We had not expected any difference in PANAS scores between different images, so we had not planned to calculate contrasts. Nevertheless, we wanted to determine what was driving the main effect of Image so we calculated pairwise contrasts for each pair of levels of image using Bonferroni adjusted alpha levels of .017 per test (.05/3). Contrasts confirmed that the reported PANAS values for the mask image, $F(1, 204) = 19.14$, $p < .001$, $\eta_p^2 = .086$, and the BLM image, $F(1, 204) = 7.28$, $p = .008$, $\eta_p^2 = .034$, were significantly higher than for the abortion protest image, but the values for the mask and BLM images, $F(1, 204) = 1.48$, $p = .22$, $\eta_p^2 = .007$, were not significantly different.

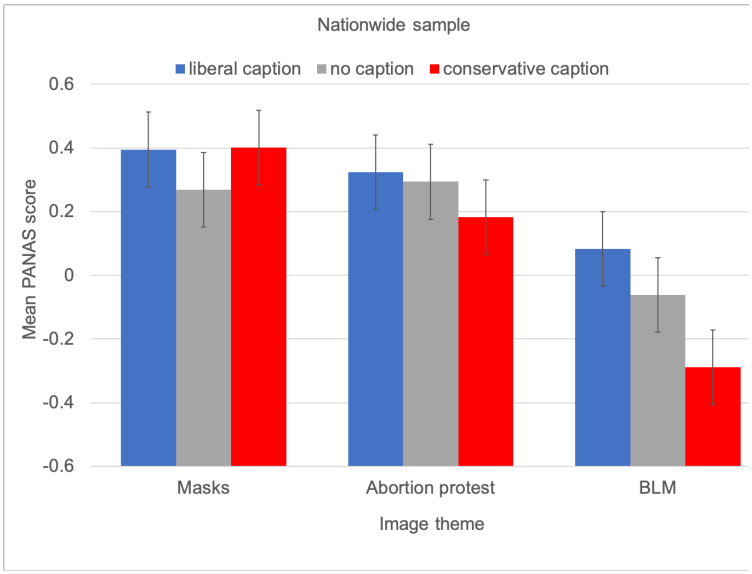
To test our first hypothesis that different captions associated with a particular image would cause different emotional responses to that image, we had planned to calculate simple effects contrasts (i.e., at a single level of the factor image) to determine the differences in PANAS scores between the liberal and conservative captions for each image using Bonferroni adjusted alpha levels of .017 per test (.05/3). The planned contrasts confirmed that the PANAS responses to the liberal caption were significantly higher than to the conservative caption only for the BLM image, $F(1, 135) = 6.49$, $p = .012$, $\eta_p^2 = .046$, but were not significantly different for the mask image, $F(1, 134) = 4.03$, $p = .047$, $\eta_p^2 = .029$, or the abortion protest image, $F(1, 135) = .62$, $p = .43$, $\eta_p^2 = .005$.

ICS. The internal consistency of the ICS was reliable ($\propto = .83$). We submitted the ICS scores from both samples to an independent samples t-test, which showed that the student sample ($M = 71.45$) were significantly more conservative than the nationwide sample ($M = 63.53$), $t(553) = 7.92$, $p < .001$, $d = .70$. This outcome was surprising, given the widespread claim expressed in the popular culture that university populations are more liberal than the overall population. One possible explanation for this surprising outcome is that our university is located in the southwestern United States, so the population from which our university's students are drawn is more conservative than the overall American population.

To test the second hypothesis that congruence between political ideology and caption would drive emotional responses, we calculated two Pearson's correlation coefficients for each image; one for the participants who were presented with the liberal caption, and one for the participants who were presented with the conservative caption. The correlation coefficients for all three images were consistent with the prediction made by our second hypothesis that the correlation coefficient for the conservative caption should be higher than for the liberal caption. For the mask image, the correlation coefficient for the conservative caption, $r(66) = .41$, $p = .001$ was greater than for the liberal caption, $r(66) = .29$, $p = .019$; for the abortion protest image, the correlation coefficient for the conservative caption, $r(66) = -.50$, $p < .001$ was greater than for the liberal caption, $r(67) = -.68$, $p < .001$; and for the BLM image, the correlation coefficient for the conservative caption, $r(67) = .39$, $p = .001$ was greater than for the liberal caption, $r(66) = .38$, $p = .001$. To see if the differences were statistically significant, we submitted the two correlation coefficients for each image to a Fisher's r to z transformation. Although the direction of the difference in correlation coefficients was consistent with our second hypothesis, none were significant, and the only one that approached significance was for the abortion protest image, $z = 1.58$, $p = .06$. For the mask image, $z = 0.77$, $p = .22$, and for the BLM image, $z = -0.07$, $p = .47$.

Figure 2

Mean PANAS responses as a function of Image and Caption for nationwide sample



Nationwide Sample

PANAS. The internal consistency of the PANAS ($\propto = .86$) was reliable. Mean PANAS responses (depicted in Figure 2) were submitted to a 3 x 3 ANOVA with Image as a within-participants factor and Caption as a between-participants factor. The main effect of Image, $F(2, 694) =$

15.68, $p < .001$, $\eta_p^2 = .043$, was significant, but the main effect of Caption, $F(2, 347) = 1.20$, $p = .30$, $\eta_p^2 = .007$, and the interaction, $F(4, 694) = 1.26$, $p = .29$, $\eta_p^2 = .007$, were not. As can be seen in Figure 2, the effect of Image appears to be driven by higher values for the mask and abortion protest images than for the BLM image.

As with the undergraduate student sample, the main effect of Image was unexpected, but we wanted to see what was driving this effect. Using Bonferroni adjusted alpha levels of .017 per test (.05/3), contrasts confirmed that the reported PANAS values for the mask image, $F(1, 349) = 45.61$, $p < .001$, $\eta_p^2 = .116$, and the abortion protest image, $F(1, 349) = 13.69$, $p < .001$, $\eta_p^2 = .038$, were significantly higher than for the BLM image, but the values for the mask and abortion protest images, $F(1, 349) = 1.05$, $p = .31$, $\eta_p^2 = .003$, were not significantly different.

To test our first hypothesis that different captions associated with a particular image would cause different emotional responses to that image, the same contrasts were planned for the differences between liberal and conservative captions as for the student sample. Using Bonferroni adjusted alpha levels of .017 per test (.05/3), only the difference for the BLM image approached significance: $F(1, 230) = 4.93$, $p = .027$, $\eta_p^2 = .021$. The PANAS responses were not significantly different for the mask image, $F(1, 240) = .002$, $p = .96$, $\eta_p^2 < .001$, or the abortion protest image, $F(1, 224) = .66$, $p = .42$, $\eta_p^2 = .003$.

ICS. The internal consistency of ICS ($\propto = .85$) was reliable. In contrast to the results from the undergraduate student sample, the correlation coefficients for the conservative captions were not higher than for the liberal captions for any of the images. For the mask image, the correlation coefficient for the conservative caption, $r(122) = .40$, $p < .001$ was less than for the liberal caption, $r(116) = .42$, $p < .001$; for the abortion protest image, the correlation coefficient for the conservative caption, $r(116) = -.47$, $p < .001$ was the same as for the liberal caption, $r(106) = -.47$, $p < .001$; and for the BLM image, the correlation coefficient for the conservative caption, $r(106) = .34$, $p < .001$ was the same as for the liberal caption, $r(122) = .34$, $p < .001$. As could be expected from such small differences between correlation coefficients, none of the Fisher's r to z transformations were significant: all ps > .05.

Discussion

Undergraduate Student Sample

Each level of the study supplies context into the relationships that may be occurring between political ideology, image and caption manipulation and emotional response, and they each have their implications and limitations that can be improved in further research. Within the undergraduate student sample we found that there were differences in PANAS scores between the image types. This finding was unexpected insofar as we believed caption type would be the factor driving an emotional response, not image type. Although the three images we selected were all politically charged, the abortion issue has more religious overtones than mask wearing and BLM, and students at a university in the southwestern U.S. can be expected to have stronger religious (and specifically Christian) affiliations than elsewhere in the country. Another possible explanation for this unexpected effect is that due to their relatively young age, university students are simultaneously more concerned with, and less experienced with, issues of reproductive health than the overall population. Thus, abortion may elicit more feelings of fear and confusion for members of our university sample than mask wearing and BLM.

Further, the political significance of abortion has radically shifted recently. At the time of data collection in late 2021 and early 2022, abortion was not as high of a political priority as it became in the summer of 2022. New developments such as the U.S. Supreme Court's suspension of the landmark Roe v. Wade decision and the subsequent defeat of a Kansas state referendum to remove abortion protection from the state constitution have propelled reproductive rights to the forefront of the nation's consciousness. Specifically, the defeat of the referendum in such a politically and religiously conservative state as Kansas suggests that the Roe v. Wade decision has inspired outspoken demands to allow women to have the right to protect their own bodily autonomy (Edelman, 2022). The fluidity of the political significance of abortion highlights one of the pitfalls associated with the kinds of questions we tried to answer in our research: political attitudes can shift suddenly and unexpectedly.

Although results from the abortion image did not support the first hypothesis claim that captions would affect emotional responses, results from the BLM image did, and the mask image came close. For the BLM image and mask image, the main effect of caption confirmed that conservative captions tended to have less positive emotional reactions than liberal captions across participants. Although we had phrased the captions with an eye to political ideology, the conservative captions emphasized conflict (mask image: "U.S. citizens ordered…", BLM: "…threatening police authority"), while the liberal captions emphasized freedom (mask: "U.S. citizens choose…", BLM: "…triumph over police power").

Although this confound between political ideology and conflict versus freedom was unintentional, it could indicate an inherent bias existing between the two ideological poles. Investigating whether the phrasing of issues for one political ideology elicits more negative emotional valence is the subject for further research. The language used in conservative captions could be inherently negative versus liberal captions, but an underlying shared belief could also be present: freedom. The liberal captions showcase either the act of freedom or the process of obtaining it while conservative captions show the opposite.

The results from images without captions did not support the first hypothesis. Instead of falling somewhere in between liberal and conservative captions, emotional responses were lower than for the captioned images. Participants are constantly exposed to information about these political subjects, so it is understandable that they would come into the study with preconceived thoughts on the political subjects. Including noncaptioned images in the study may have been a limitation which could be eliminated by omitting them from future studies.

National Sample

With a sample of a much greater spread of diversity we can see a stark difference in the results of the study. As with the university students, one of the images elicited emotional responses with more negative valence than the other two images, but for the national sample it was the BLM image rather than the abortion protest image. As with the student sample, this image may have more immediacy for the community sample than the other two images. In contrast to the student sample for whom reproductive issues elicit more negative emotions, the community sample may be more sensitive to issues of public safety. The noncaptioned images supported the first hypothesis more for the national sample than the student sample. For the BLM and abortion images, emotional responses to the noncaptioned images was between the response to the liberal and conservative captions, but not for the mask image. As with the student sample, the BLM image elicited higher positive emotional response to liberal captions than conservative captions. A possible explanation for the null effect of the mask image for the national sample, in relation to undergraduate student sample, is that the national sample was collected after the undergraduate student sample, so wearing masks had become less of a political matter and more of a public health matter. Therefore, the participants from the national sample may have had less polarized views on the topic than the undergraduate students.

The BLM image is the only image that elicited general agreement across the two samples, with more positive emotional reactions for liberal captions and less positive emotional reactions for conservative captions. We believe this may be due to the different ways people confront these subjects. Abortion is a subject that is political, but also has religious significance. Wearing masks is an issue of public health and safety that became a political subject in a time of intense public health threat. The BLM movement is the only overtly political movement of the three as it focuses on political and social rights of a group of people. It would be interesting to take this study and narrow down subjects to raw political significance to see if the same relationship occurs as the BLM images.

While there was some agreement between the two samples in their support of the first hypothesis, both samples failed to support the second hypothesis. For the undergraduate students sample, although results from all three images were consistent with the second hypothesis claim that relatively liberal participants would have a more positive response to liberal captions, and more negative response to conservative captions, than relatively conservative participants, none of these results reached significance. For the national sample, results from none of the three images supported this hypothesis, and again, none of the results reached significance.

Generalizability of Student Samples

As our results indicate, there was modest agreement between the student and national samples. This led us to consider how our study extends on previous work examining the generalizability of convenience samples of university students to community samples. Hanel and Vione (2016) gathered personality and attitudinal responses from 59 countries, and found discrepancies between the student and community sample. They concluded that generalizing from a student sample may be problematic for a study such as theirs that examined personality and attitude variables. On the other hand, Smith et.al. (2022) looked at the classic asymmetry between the Stroop (1935) and reverse Stroop effects, which reflect differences in basic perceptual and cognitive processing. The Stroop and reverse Stroop effects were consistent between the student sample and national sample.

Because Smith et al. (2022) focused on basic perceptual and cognitive processing, their questions had objectively right and wrong answers. When a participant in Smith et al. was asked to report the pixel color of the word “Red” written in blue pixels, the correct answer was “Blue,” not “Red.” As a result, when participants provided an incorrect response, the script could provide feedback indicating an incorrect response, and the experimenters could exclude the incorrect responses from their

analyses. As in Hanel and Vione (2016), our study focused on variables that reflect subjective quantities (e.g., personality, attitudes, emotions), and as a result we could not determine if participants were providing the correct answers.

Thus it seems that our results contribute to the question about whether the results from convenience samples of students generalize to the overall population. Perhaps for experiments that examine basic perceptual and cognitive processing, and ask questions for which only one answer is objectively correct (as in Smith et al., 2022), the results from convenience and community samples might be expected to enjoy favorable agreement. On the other hand, for experiments that examine higher level cognitive processing, and ask participants to report their personal subjective experiences (as in our experiment, and in Hanel & Vione, 2016), the agreement between the convenience and community samples might be expected to be modest. We argue that a key tool in addressing the question of whether the results from convenience samples of students generalize to the overall population would be experiments such as ours that present identical materials to both kinds of samples.

Limitations and Future Directions

Although our study contributes to the ongoing discussion of the generalizability of convenience samples, it also has limitations that we should acknowledge. First, all participants were exposed to the three images in the same order: the mask image, then the abortion protesters, then the confrontation between BLM and police. We manipulated the captions between participants with the expectation that any differences in emotional responses to the images would be attributable to the captions. But the fixed presentation order between participants might have introduced carryover effects from previous images that are confounded with the caption effects. Second, the captions we wrote may have failed to elicit the emotional reactions we had intended. And even worse, while a caption such as "Protestors conform towards taking rights from women" seemed clear to us when we wrote it, in retrospect its interpretation is not as clear as we intended. Perhaps the main defect with our captions is their brevity, which could have obscured their meaning. Of course, when we developed our materials, we deliberately wrote brief captions in the hope that participants would be more likely to read them than if the accompanying text were longer.

While the primary contribution of our study is what it adds to the discussion about the generalizability of the results from convenience samples, the limitations we identify here reduce its importance. Nevertheless, a future study could build on ours by exploiting its strengths while redressing its defects. As we argued above, the key to determining what kinds of experimental results generalize from convenience samples would be to deliver identical experimental materials to both kinds of samples. A study that looks at the effects of captions on emotional reactions should present the images in random order to eliminate the confounding order effects. The accompanying text should be longer than the brief but ambiguous captions we created. Then, to ensure that participants actually read the longer text that accompanies the images, the experimental script could ask participants to respond to questions that have objectively correct answers. Ultimately, if researchers compare the results from convenience and community samples across a diverse swath of experimental methods, it may reveal what kinds of experimental methods generalize from convenience samples, thereby validating the widespread use of college sophomores as participants in psychological science.

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Appendix A
Stimulus Conditions for Survey

This appendix shows the images and caption stimuli for the three conditions in the survey.

Condition One	Condition Two	Condition Three
U.S. citizens choose to wear face masks for public safety.		U.S. citizens ordered to wear face masks for COVID protection.
Protesters work towards accomplishing rights for unborn children.	Protesters conform towards taking rights from women.	
	BLM protesters threaten police authority.	BLM protesters triumph over police power.

Appendix B
Ideological Consistencies Scale statements

Conservative	Liberal
Government is almost always wasteful and inefficient.	Government often does a better job than people give it credit for.
Government regulations of business usually does more harm than good.	Government regulation of business is necessary to protect public interest.
Poor people today have it easy because they can get government benefits without doing anything in return.	Poor people have hard lives because government benefits don't go far enough to help them live decently.
Blacks who can't get ahead in this country are mostly responsible for their own condition.	Racial discrimination is the main reason why many black people can't get ahead these days.
Immigrants today are a burden on our country because they take our jobs, housing and healthcare.	Immigrants today strengthen our country because of their hard work and talents.
The best way to ensure peace is through military strength.	Good diplomacy is the best way to ensure peace.
Most corporations make a fair and reasonable amount of profit.	Business corporations make too much profit.
Stricter environmental laws and regulations cost too many jobs and hurt the economy.	Stricter environmental laws and regulations are worth the cost.
Homosexuality should be discouraged by society.	Homosexuality should be accepted by society.
The government today can't afford to do much more to help the needy.	The government should do more to help needy Americans even if it means going deeper into debt.